2024 HCI Prelim H1 Geography Marking Guide

1(a)	Explain why the hypothesis is appropriate for the investigation. [3] Award 1 mark for each explanation of appropriateness of the hypothesis to a maximum of 3 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses include: • Suitable scale ○ temporal scale – one month is sufficient time to collect data from the targeted sample size [1] ○ Spatial scale – students have access to the target population 'consumers in Singapore' [1] • Researchable ○ data can be collected via surveys in the study of consumers' perceptions [1] • Clearly defined ○ Independent variables such as community, government and businesses were stated [1] ○ Dependent variable - importance of role of actor – was also included in the hypothesis [additional 1 m]
(b)	With reference to Resource 1, suggest the challenges that the students may face conducting the survey online and at Bugis Junction. [4] Award 1 mark for each suggestion on the challenges in conducting the surveys, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable. Possible responses include: Online • Lack of responses as many people may choose to ignore the message [1] • Respondents may not complete the survey accurately [1] Bugis Junction • Unwillingness of the public to do the survey [1] • Lack of passers-by in the afternoon [1] • Passers-by who are rude and dismissive and may act aggressively when approached [1] Both • Language barrier if the survey is only in English. Potential survey participants may only be able to communicate in Chinese dialects, Malay or Tamil. [1]
(c)	With reference to Resource 1, suggest how the students could minimise potential risks when administering the survey at Bugis Junction. [5] Award 1 mark for each suggestion on minimising potential risks, to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable. Possible responses include: • High temperature in the afternoon (stated that they conducted the survey in the afternoon)

	○ Students should conduct the survey in the evenings where possible when the temperatures are lower [1] ○ Stay well hydrated and take breaks as without proper hydration or rest, students may faint and injure themselves or put themselves in a dangerous situation e.g. collapsing when crossing the road [1] • Crossing of roads ○ Be aware of road crossings at the carpark entrances and check for traffic before crossing as vehicles can move very fast and drivers may not see the pedestrians [1] ○ Do not jaywalk, follow all road signs and check for traffic before crossing the road [1] • Conducting the survey individually ○ Students should work in (at least) pairs to help seek help when an emergency occurs [1] ○ Students should stay in areas where there is more human traffic and avoid quiet and isolated areas [1]
(d)	Explain the ethical considerations that the students should take note of when conducting the interviews with family members and neighbours. [4] Award 1 mark for each explanation of ethical considerations for interviews to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible responses include: • The students should not interrupt the interviewees in their daily activities [1]. They could make an appointment with the interviewees first, ensuring that they can make time and are available for the interview [+1]. • Informed consent must be obtained, typically in written form, ensuring that participants understand what their involvement entails [1]. Participants should be fully informed about the purpose, procedures, and potential risks of the study before they agree to take part [+1]. The information should be presented in clear, accessible language, avoiding jargon or technical terms [+1]. Participants should be aware that their participation is voluntary, and they can withdraw at any time without penalty [+1]. • Researchers must protect the privacy of participants by ensuring that their personal information and responses are kept confidential [1]. Data should be made anonymous where possible, with the removal of names or other identifying details replaced with pseudonyms [+1]. The researcher must ensure that the collected data is stored securely, whether in physical or digital form, to prevent unauthorized access [+1]. • Participants should be treated with respect, and their dignity must be upheld throughout the research process [1]. The students should be sensitive to cultural, social, and personal differences and avoid questions or behaviour that could be considered intrusive, offensive, or disrespectful [+1]. The students must take care not to cause physical, psychological, or emotional harm to participants [+1]. This includes being mindful of sensitive topics that could trigger distress [+1]. • The students must be aware of potential power imbalances between themselves and participants, especially in contexts involving vulnerable groups (e.g., elderly, children, marginalized communities) [1]. Participation must be entirely voluntary, and there should be no pressure or coercion for individuals to take part [+1].
(e)	Compare the consumers' perception of the actors' importance in fighting climate change shown in Resource 2. [4] Award 1 mark for each comparison to a maximum of 4 marks.

	Award a maximum of 1 additional mark for further development of each comparison, where applicable. Possible responses include: Similarities • Strongly disagree and category 2 is the lowest for all actors. All 5% or less. [1] • Strongly agree and the category 5 together form the highest % for all and they are all above 55% [1] • Government and community groups have the same value for Strongly agree and cat 5 at 68% [1] Differences • Businesses rank the highest at 71% for Strongly agree and category 5 which is higher than the next highest values which are for Government and Community groups at 68% [1] • However, Government is ranked the highest for Strongly agree at 38% amongst all the actors followed rather closely by Businesses at 34% and Community groups at 32% [1]
(f)	The students concluded that consumers in Singapore think that community response to climate change is more important than the government and businesses. With reference to Resources 2 and 3 evaluate the validity of their conclusion. [10] Possible Approaches: Candidates could refer to Resources 2 and 3 and consider how accurate and reliable the collected data are for drawing conclusion about the importance of community, government and business responses to climate change. Candidates could look at the way data are collected and assess if the methods used could make accuracy and reliability of the data questionable. For instance, they could look at the sample size, the use of online surveys, sampling method at the shopping mall and sampling method for the interviews. Information from Resources 2 and 3 could be used to support or disprove the conclusion made by the students. Additionally, candidates could consider what other data would be needed for this investigation. For instance, they could mention that the study could analysis responses by gender, age groups, education level and income level to gain a deeper understanding of consumer perception. Levels marked using Generic Level Descriptors for H2 fieldwork evaluative questions

2 (a)	Cite data to compare urban growth in selected cities shown in Resource 4. [3] Award 1 mark for each comparison of urban growth in selected cities, to a maximum of 3 marks. Award a maximum of 1 additional mark for further development of each description, where applicable. Award a maximum of 2 marks for comparison of either similarities or differences only. Similarity • All cities, except London, experienced urban growth. E.g. New York 12 million in 1950 to 20 million in 2025 • Cities such as Lagos, Bombay and Cairo have the greatest increase in urban growth between 2000 and 2025 e.g. Bombay's 10 million increase Difference • Tokyo has the highest urban growth with 25 million increase between 1950 and 2025, significantly much higher than other DC cities such as New York with only 8 million increase

	• London is the only city with a decrease. There was a fall between 1950 and 1975 of 1%.
(b)	With reference to Resource 4, explain how demand may be placed on natural environments in Lagos and Mumbai. [4] Award 1 mark for each explanation of demand placed on natural environments, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. High concentration of waste • High urban growth leads to high generation of waste. This could come from industrial, commercial and domestic sources. Waste in LDC cities is often dumped on open ground and this can cause land, water and air pollution. Forests may be cleared for such purposes and the pollution affects natural plant growth and the health of aquatic wildlife if the toxins leach into the groundwater and into rivers. Large ecological footprint • Population growth places extraordinary demand for natural resources and exceptional stress on natural systems. For example, over 13 million hectares of forest land are converted to agriculture, urban land use, and industrial forestry annually. This deforestation significantly affects both hydrologic systems and territorial habitats. • Ecologically, these stressors significantly affect aquatic flora and fauna. Habitat loss is the greatest threat to biodiversity. Urbanization not only destroys and fragments habitats but also alters the environment itself. For example, deforestation and fragmentation of forest lands lead to the degradation and loss of forest interior habitat as well as creating forest edge habitat. Vast quantities of resources absorbed by urban areas from the surrounding areas • Extraction of raw materials to supply to the economic activities or domestic needs of the city. Raw materials such as wood and metals are extracted from the natural environment to supply materials needed for the production process in the manufacturing industry. In addition, food sources are dependent on natural environment to reproduce and • Natural resources are also extracted from the natural environment to meet the demand for energy. Non-renewable resources such as fossil fuels like coal and oil are extracted to be used in industries and to run motor vehicles. The combustion of these fossil fuels release greenhouse gases into the atmosphere, increasing their concentration and causing global warming which in turn leads to loss of natural habitats and biodiversity. Other demands • Hydrologically, urbanization creates a condition called urban stream syndrome. The increase in storm runoff, caused by urbanization through the addition of impervious surfaces, alters stream flow, morphology, temperature, and water quantity and quality.
(c)	With reference to Resource 5, suggest the possible challenges faced by New York and Boston in progressing towards sustainable urban development. [4] Award 1 mark for each suggestion on the possible challenges faced by NY and Boston, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable. Challenges of urban population loss • Economic decline and urban dereliction:

	- As the urban area loses its population, land in some areas and neighborhoods may lose their value, resulting in zones of abandonment where empty buildings often fall prey to accumulating garbage, crime, and vandalism. [1] For example, due to industrial decay, cities such as Boston and New York may suffer from abandonment of many properties. This poses challenges to sustainable urban development as increased social polarization and crime rates pose social and the associated economic issues. [1] • A shrinking workforce due to urban population loss can lead to labour shortages, or make the city less attractive for business owners / investors. Businesses can therefore leave the city, leading to a loss of jobs and deindustrialisation. [1] • The ageing and decay of buildings with the passage of time, especially with businesses and residents relocating in search of better job opportunities. [1] • Social Impacts of deindustrialisation in urban areas include an increase in unemployment, higher levels of social issues such as crime, drug abuse and family breakdown, and the out migration of skilled population. [1] This has an economic impact too, with a de-multiplier effect, decline in property prices, loss of jobs, lower disposable incomes, loss of local tax base. [1] • The environment of central areas (urban environment) can suffer too, with dereliction of buildings, land pollution and reduced housing maintenance. [1] In addition, there can be a sizeable supply of brownfield land in places such as Boston and New York. Brownfield sites are parcels of land that was previously used for industrial purposes and which are contaminated by low concentrations of hazardous chemicals. [1] • In addition, as cities shrink, their governments have to maintain infrastructure built for a larger population with smaller tax revenues. [1] Other problems may arise. For example, lower demand for water means that water may stay in distribution pipes for longer periods, increasing the risk of bacterial growth and contamination. Hence, shrinking cities need to find effective ways to downsize existing infrastructure and find a usable financial model for its operation and maintenance. [1]
(d)	With reference to Resource 6, explain the limitations of landfills as a waste management strategy. [6] Award 1 mark for each suggestion on the limitations of landfills, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable. • Decomposition of waste materials cause the release of methane which is flammable and when fires start on landfills, they are uncontrollable and tend to be huge. [1] • Landfills generate leachate, a toxic liquid formed when rainwater filters through waste. If not properly managed, leachate can seep into soil and groundwater, leading to contamination of water sources and harming ecosystems. [1] • Decomposing waste in landfills emits harmful gases, particularly methane, a potent greenhouse gas that contributes to climate change. Landfills also release volatile organic compounds (VOCs) and other pollutants, which can degrade air quality. [1] • Landfills require large areas, which can lead to the loss of valuable land. As cities expand, finding new sites for landfills becomes increasingly difficult, especially in densely populated regions. [1] • Landfills have limited capacity and eventually reach their maximum limits. Once full, new landfill sites must be found, leading to a continuous need for more space. [1] • Landfills can pose health risks to nearby communities due to the emission of toxic substances, which may include heavy metals, chemicals, and pathogens. Residents near landfills may experience higher rates of respiratory problems, skin irritations, and other health issues. [1] • Landfills can attract pests such as rats, flies, and other vermin, which can spread diseases and create unhygienic conditions. [1]

	• Many items disposed of in landfills, such as plastics, do not decompose for hundreds of years, leading to long-term environmental damage. Even organic materials break down slowly in landfill conditions due to limited oxygen and other factors. [1] • Organic waste that decomposes in landfills could otherwise be used for composting or energy generation (e.g., through anaerobic digestion). Disposing of organic waste in landfills wastes these potential resources. [1] • The "Not In My Backyard" (NIMBY) phenomenon often leads to strong public opposition (community resistance) to new landfill sites due to concerns over environmental pollution, decreased property values, and health risks. [1] • Landfills are often located near low-income or marginalized communities, leading to environmental injustice as these populations disproportionately bear the negative effects. [1] • Landfills are often unsightly and can negatively impact the surrounding landscape, reducing the aesthetic appeal of an area and lowering property values. [1] • Landfills have a limited operational lifespan and eventually close, leading to the need for new sites. Relocating and constructing new landfills is not only costly but also increasingly difficult due to space limitations and public resistance. [1]
(e)	With reference to Resource 7, suggest how less demand is placed on the environment when waste is viewed as a potential resource. [5] Award 1 mark for each suggestion on lessened demand on the environment. Award a maximum of 1 additional mark for further development of each suggestion, where applicable. From Resource 7 • When solid waste can be converted biogas, heat and electricity for industrial and domestic use, less natural resources such as coal and oil (or even alternative energy) is needed to generate energy to meet the demand for energy in cities. Hence, less natural resources are extracted from the natural environment. [1] ○ The production of energy using fossil fuel also produce greenhouse gases and air pollution which would put stress on the environment. Thus, using less of such resources would put less stress on the environment. [1] • Green jobs are generated in such multi-utility energy companies which might divert employment from other less sustainable forms of employment such as forestry and energy plants using fossil fuels. [1] Other content • Materials that can be recycled and reused would reduce the need to extract resources to produce goods and less waste is also generated which reduces pollution of the natural environment. [1]

3(a)	Cite data from Resource 8 to describe the changes in global surface temperature for annual average and natural drivers only between 1880 and 2020. [4]
	Award 1 mark for each description to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable. • Overall trend is that the annual average has increased (change of 1.2°C) while natural drivers only has made no changes (0°C) between 1880 to 2020 • Both annual average and natural drivers only started at almost the same temperature change in 1880 (0.2 and 0.1°C respectively)

	• Annual average experienced an overall no increase in temperature between 1880 – 1970 but between 1970 – 2020, it rose steeply/rapidly by 1.1°C • The annual average has clear fluctuations with about 4 peaks per 20 years while natural drivers only has fewer and less distinct fluctuations • Both annual average and natural drivers only have the same lowest temperature increase at -0.3°C but in different years (1910 and 1990 respectively)
3(b)	With reference to Resource 9, compare the effect of habitat change and climate change on biodiversity in ecosystems. [5]
	Award 1 mark for each comparison to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each comparison, where applicable. • Overall habitat change has a greater impact on biodiversity than climate change [1] ○ The greatest impact at 'very high' is experienced in ecosystems such as tropical forests and temperate grassland for habitat change, but none of the impacts are classified as 'very high' for climate change [1] ○ 61% (8/13 ecosystems) of the impact of climate change falls under 'low' impact while only 23% (3/13 ecosystems) of the impact of habitat change fall under 'low' impact [1] • The current trends of the impact of habitat change cover all the 4 different current trends but all of the impact of climate change' current trend is classified as 'very rapid increase' [1] ○ However, 9/13 or 69% of the impact of habitat change current trends is classified as 'increasing' and 'very rapid increase' of impact. [1]
3(c)	Explain how ecosystems may be affected by contemporary climate change. [4]
	Award 1 mark for each explanation to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Impacts on Aquatic Ecosystems • Warmer water temperatures can alter the metabolism, reproduction, and growth rates of aquatic organisms. Species like corals, fish, and phytoplankton are particularly sensitive to temperature changes. • Coral bleaching is a direct result of rising sea temperatures, as corals expel the symbiotic algae living in their tissues, which leads to a decline in coral reefs. • Increased carbon dioxide (CO_2) absorption by oceans is lowering the pH of seawater, making it more acidic (ocean acidification). This affects organisms that rely on calcium carbonate to build shells or skeletons, such as corals, mollusks, and some plankton species. Acidification can disrupt the marine food chain, leading to declines in species that depend on these organisms. • Climate change is disrupting ocean currents, which play a crucial role in regulating global climate and nutrient distribution. Shifts in circulation patterns can affect marine ecosystems, altering the availability of nutrients for organisms and potentially causing shifts in fish populations. • Rising sea levels are threatening coastal habitats such as mangroves, salt marshes, and wetlands. These ecosystems act as buffers against storm surges and provide critical habitats for many species. The loss of these areas can reduce biodiversity and increase the vulnerability of coastal communities to natural disasters. • Changes in precipitation patterns, increasing evaporation, and altered snowmelt regimes are affecting freshwater availability. Lakes, rivers, and wetlands are experiencing changes in water levels, temperature, and quality, affecting species like fish, amphibians, and aquatic plants. • In many regions, droughts and reduced freshwater flow are causing habitat degradation and threatening species that rely on specific water conditions. • Warmer waters are driving species to migrate toward cooler areas, often leading to shifts in the distribution of fish populations. For instance, commercially important species like cod and tuna are moving to different latitudes or depths in search of suitable habitats.

	Impacts on Terrestrial Ecosystems: • Changes in temperature, precipitation, and extreme weather events (such as droughts, floods, and storms) are directly affecting plant growth, water availability, and ecosystem productivity. This leads to shifts in vegetation zones and affects species that depend on specific habitats. • Many terrestrial species are shifting their ranges toward higher altitudes or latitudes in response to warming temperatures. This can lead to changes in community composition, where certain species outcompete others, potentially driving some to extinction. • Arctic and alpine ecosystems are particularly vulnerable as species have less room to migrate and are losing critical habitats. • Climate change is causing shifts in phenology, which refers to the timing of biological events like flowering, migration, and reproduction. Warmer temperatures are leading to earlier blooming of plants, earlier animal migrations, and changes in breeding seasons. These shifts can lead to mismatches in timing between species, such as when plants bloom before pollinators emerge, disrupting ecosystem interactions. Impacts on both • Climate change is accelerating species extinction, particularly for those unable to migrate or adapt quickly enough. Sensitive ecosystems like tropical rainforests, tundra, and coral reefs are experiencing significant biodiversity losses, which in turn impacts ecosystem functioning and resilience. For example, polar bears in the Arctic are losing access to sea ice, which is critical for hunting, while amphibians are especially vulnerable to temperature and moisture changes in their habitats.
3(d)	With reference to Resource 10, describe the main effects of global warming on the Earth's climate systems. [4]
	Award 1 mark for each explanation to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. • Overall, the impact of global warming on tipping points of climate systems are found on both land and sea but are located to a larger extent in the Northern Hemisphere. • The locations that global warming have the fastest impacts are the Greenland ice sheets, Labrador-Irminger Seas Convection and West Antarctic Ice sheet where the impacts are global in nature with tipping points at the lowest temperature increases (1.5 - <2.0°C). o Those with impacts that would take longer to come into effect are those with the greatest temperature increases of >6°C e.g. East Antarctic Ice Sheet and the Arctic Winter Sea Ice • The locations where global warming impacts take the longest to come into effect are regional in nature with tipping points at the higher temperature increases of 3.7 – 6.0°C such as the Northern Forest. o Among those with regional elements, those that take the shortest time to come into effect are the Low Latitude Coral Reefs, Boreal Permafrost and Barents Sea Ice. • Out of 9 climate systems with global elements, 4 involve permafrost, sea ice or ice sheets e.g. Boreal permafrost
3(e)	With reference to Resources 8, 9 and 10, suggest how changes to aquatic and terrestrial ecosystems can impact humans. [5]
	Award 1 mark for each suggestion to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable. • Rising sea temperatures, ocean acidification, and changing marine ecosystems are leading to a decline in fish stocks. Many coastal and island communities depend heavily on fish as a primary protein source. Changes in fish migration patterns or declines in populations affect the availability of food and the livelihoods of millions. [R8, R9, R10] • Changes in terrestrial ecosystems, including altered precipitation patterns, increased frequency of droughts, and changes in temperature, affect agricultural productivity. Crops that are sensitive to heat stress, like wheat, rice, and maize, may experience reduced yields,

	threatening food security for billions of people globally, particularly in vulnerable regions. [R8, R9] • Changing precipitation patterns and melting glaciers are affecting freshwater availability in many regions. Rivers fed by glaciers are seeing reduced flows, impacting water supply for agriculture, industry, and drinking water in areas like South Asia and the Andes. [R8, R9, R10] • Coastal communities are particularly vulnerable to sea-level rise and increased frequency of extreme weather events like hurricanes and storm surges. Infrastructure, such as homes, roads, and industries, faces a higher risk of damage, leading to costly repairs and relocation efforts. [link to rising temperature and melting ice leading to sea level rise in R8 and R10] • As ecosystems become less hospitable due to rising sea levels, desertification, and increased disasters, millions of people may be forced to migrate, creating climate refugees. This can lead to increased pressure on urban areas, competition for resources, and potential conflicts. [R8, R9]

4.	'Successful urban reimaging strategies must include consideration of sustainable urban development'. Evaluate this statement.	[13]
	Possible approaches: Candidates could approach the question by making a judgement on whether urban reimaging strategies that achieve success must take sustainable urban development into consideration through the use of two or more case studies. Candidates could evaluate based on - Whether the strategies in the case study achieved the intended aims/targets and included elements of SUD - Whether there are unintended negative impacts in achieving aims/targets even when SUD is considered - Whether other factors such as good governance, inclusion of local participation in planning of strategies play a role in the success of strategies - Challenges to success across case studies - Extent of success based on economic scale, temporal scale, progress towards SUD	
5.	"Given the rate of contemporary climate change, mitigation measures are more urgent than adaptation measures." Evaluate this statement.	[13]
	Possible approaches: Candidates could approach the question by making a judgment on whether mitigation measures are more urgent than adaptation measures. Candidates could evaluate based on - Certain contexts/countries in which mitigation measures might be more urgent than adaptation measures, and vice versa. - Relative urgency in terms of economic, temporal or spatial scale.	
6.	Evaluate the influence of human activities on climate change.	[13]
	Possible approaches: Candidates could approach the question by making a judgement on whether human activities or natural factors have a greater influence on climate change. Candidates could evaluate based on	

	- Influence of human activities/natural factors over time and spatial scale - Effectiveness of strategies and/or governance, which affects the extent of impact of human activities.	